Solutions to MP204 Problem Set 2

(1)

P.1

Despite the fact that this looks different, we've effectively already done this in lecture: recall that we showed that for we have a uniformly-charged solid sphere of total charge Q and radius R, the electric field was given by

$$\vec{E} = \begin{cases} \frac{1}{4\pi\epsilon_0} \frac{Q}{R^3} \vec{r} & |\vec{r}| < R \\ \frac{1}{4\pi\epsilon_0} \frac{Q}{|\vec{r}|^3} \vec{r} & |\vec{r}| > R \end{cases}$$

We obtained this using $\oint_S \vec{E} \cdot d\vec{S} = Q_S/\epsilon_0$. But a simple substitution turns Gauss' Thm for electric fields into one for gravitational fields:

$$\vec{E} \to \vec{g}, \quad Q_S \to M_S, \quad \frac{1}{\epsilon_0} \to -4\pi G$$

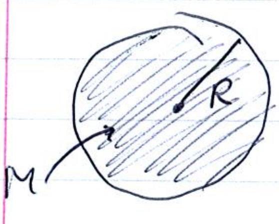
so that

$$\oint_S \vec{E} \cdot d\vec{S} = Q_S/\epsilon_0 \longrightarrow \oint_S \vec{g} \cdot d\vec{S} = -4\pi G M_S.$$

Thus, with these substitutions, we find

$$\vec{g} = \begin{cases} \left(\frac{-4\pi G}{4\pi}\right) \frac{(M)}{R^3} \vec{r} = -\frac{GM}{R^3} \vec{r} & |\vec{r}| < R \\ \left(\frac{-4\pi G}{4\pi}\right) \frac{(M)}{|\vec{r}|^3} \vec{r} = -\frac{GM}{|\vec{r}|^3} \vec{r} & |\vec{r}| > R \end{cases}$$

for the gravitational field due to a uniform sphere of mass M. But if you don't like this explanation, let's do it in full...



What's the mass density of this sphere? Well, if the total mass M is uniformly distributed, then the density is

$$\rho = \frac{\text{total mass}}{\text{volume}} = \frac{M}{\frac{4}{3}\pi R^3} = \frac{3M}{4\pi R^3}$$

Now, let S be a spherical surface of radius r centred on the sphere; what's the mass M_S contained within S? If $r > R$, then S contains the entire sphere, and so the mass inside S is simply the total mass M. If $r < R$, however, S contains only a part of the mass. However, we can easily compute it:

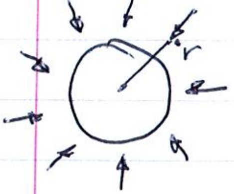
$$M_S = \int_{r<R} \rho \, dV = \int_0^r \frac{3M}{4\pi R^3} \cdot 4\pi (r')^2 dr'$$

$$= \frac{3M}{R^3}\int_0^r (r')^2 dr' = \frac{3M}{R^3}\left(\frac{1}{3}r^3\right) = \frac{r^3}{R^3}M,$$

where we used $dV = 4\pi(r')^2 dr'$ as the volume of a spherical shell of radius r' and thickness dr'. Thus,

$$M_S = \begin{cases} \frac{r^3}{R^3}M & r<R \\ M & r>R. \end{cases}$$

Now, the symmetry of the case indicates that the gravitational field $\vec{g}$ can only point in the radial direction, regardless



of whether or not we're inside the sphere. Further, $|\vec{g}|$ can depend only on the distance r from the sphere's centre. Therefore, $\vec{g} = g(r)\hat{r}$ is the general form of the gravitational field.

Now, if S' is a spherical surface centred on the sphere, then $d\vec{S}$ points straight out, i.e. $d\vec{S} = |d\vec{S}|\hat{r}$. Therefore, $\vec{g}\cdot d\vec{S} = g(r)|d\vec{S}|$, and so $\oint_S \vec{g}\cdot d\vec{S} = \oint_S g(r)|d\vec{S}|$.

But r is constant on S', so $g(r)$ has the same value everywhere, and can thus be pulled outside of the integral: $\oint_S \vec{g}\cdot d\vec{S} = g(r)\oint_S |d\vec{S}|$. $\oint_S |d\vec{S}|$ is just the sum of all the little bits of surface elements, and is thus the total surface area of a sphere of radius r, i.e. $4\pi r^2$. Therefore, $\oint_S \vec{g}\cdot d\vec{S} = 4\pi r^2 g(r)$.

This new version of Gauss' Theorem thus gives

$4\pi r^2 g(r) = -4\pi G M_S$, or $g(r) = -\frac{GM_S}{r^2}$.

Putting in $\hat{r} = \vec{r}/|\vec{r}| = \vec{r}/r$, we have $\vec{g} = -\frac{GM_S\vec{r}}{r^3}$.

But we know M_S from above, and so putting it in gives

$$\boxed{\vec{g} = \begin{cases} -\frac{GM\vec{r}}{R^3} & r<R \\ -\frac{GM\vec{r}}{r^3} & r>R \end{cases}}$$

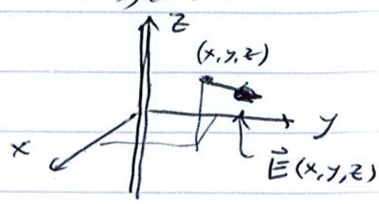
(Note that we get the same result as in the electric case: from the outside, a uniform sphere of mass looks like a <u>point</u> mass no matter how big it is. Inside, however, its radius R <u>does</u> matter.)

P. 2

Let's rewrite $\vec{E}$ in Cartesian coordinates as the hint suggests:

$$\vec{E}(x,y,z) = \frac{\lambda}{2\pi\epsilon_0}\left(\frac{1}{\sqrt{x^2+y^2}}\right)\left(\frac{x\hat{i}+y\hat{j}}{\sqrt{x^2+y^2}}\right)$$

$$= \frac{\lambda}{2\pi\epsilon_0}\,\frac{x\hat{i}+y\hat{j}}{x^2+y^2}$$



or

$$E_x = \frac{\lambda}{2\pi\epsilon_0}\frac{x}{x^2+y^2},\quad E_y = \frac{\lambda}{2\pi\epsilon_0}\frac{y}{x^2+y^2},\quad E_z = 0.$$

Now, at $r=0$, $x=y=0$, and these expressions aren't defined. Thus, all calculations are done assuming $r>0$, including computing the divergence:

$$\vec{\nabla}\cdot\vec{E} = \frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z}$$

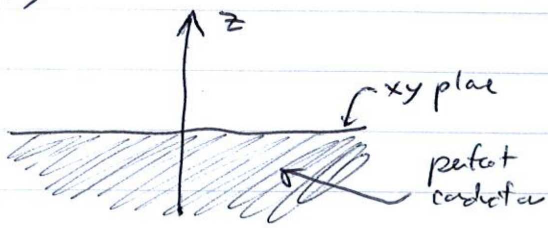
$$= \frac{\lambda}{2\pi\epsilon_0}\left[\frac{\partial}{\partial x}\left(\frac{x}{x^2+y^2}\right) + \frac{\partial}{\partial y}\left(\frac{y}{x^2+y^2}\right)\right]$$

$$= \frac{\lambda}{2\pi\epsilon_0}\left[\left(\frac{1\cdot(x^2+y^2)-x(2x)}{(x^2+y^2)^2}\right) + \left(\frac{1\cdot(x^2+y^2)-y(2y)}{(x^2+y^2)^2}\right)\right]$$

$$= \frac{\lambda}{2\pi\epsilon_0}\left[\frac{y^2-x^2}{(x^2+y^2)^2} + \frac{x^2-y^2}{(x^2+y^2)^2}\right] \boxed{= 0}$$

so the divergence of this electric field does indeed vanish. Why did we expect this? Recall that Maxwell's first eq'n has two forms: the integral form $\oint_S \vec{E}\cdot d\vec{S} = Q_S/\epsilon_0$ and the differential form $\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0$. We got this electric field in lecture using the integral form, but it also must satisfy the differential form. Luckily, it does: all the charge in the system is purely on the wire, which means that the density is only nonzero along the z-axis. In other words, $\rho = 0$ for $r>0$. But this is exactly what $\vec{\nabla}\cdot\vec{E} = 0$ says: $\vec{\nabla}\cdot\vec{E} = \rho/\epsilon_0$ vanishes only if $\rho = 0$. $\boxed{\text{Thus, we expected } \vec{\nabla}\cdot\vec{E} = 0.}$
(In fact, we would have had cause to worry if $\vec{\nabla}\cdot\vec{E}$ wasn't zero!)

P.3



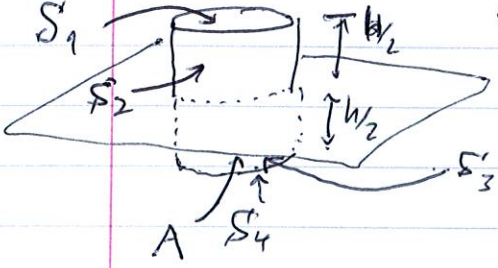
We have a perfect conductor filling all of the lower half of space in this problem, as illustrated to the left. The question is, what is the electric field everywhere in space?

The fact that it is a perfect conductor gives us half the answer: because all the charges push away from each other, and they are completely free to move, they all end up on the surface of the conductor, i.e. we have a surface charge density σ at $z=0$, and $\vec{E}=0$ inside the conductor ($z<0$). If $\vec{E}$ were nonzero, then the charges would still feel forces, and thus would continue moving, but that can't happen once they're all wellspread out to the surface.

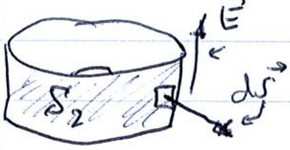
Now, this system is "invariant" under shifts in the x- or y-direction. This ~~means~~ that any push of the system in a horizontal direction leaves it looking exactly the same. Thus, none of the physical quantities of the system — in this case, the surface charge density σ and the electric field $\vec{E}$ — can depend on either x or y. Thus, σ is constant and $|\vec{E}|$ can only depend on z. Furthermore, this symmetry also implies $\vec{E}$ can only have a z-component. We can use the same argument as in lecture: a ring of charge on the

surface will give bits of electric fields to the point directly above its centre, but the sum of opposite bits will sum to a vector only pointing straight up. Thus, $\vec{E} = E(z)\,\hat{k}$ is the only possible form of the electric field.

Now we want to use $\oint_S \vec{E}\cdot d\vec{S} = Q_S/\epsilon_0$; what surface S should we use? We want one that actually contains charge (so that $Q_S \neq 0$), and so our surface must intersect the $z=0$ plane, where all the charge is. The symmetry of the system suggests a small cylinder intersecting the plane is a good choice, as shown below. Let's say the cross-sectional area of

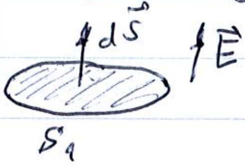


the cylinder is A and it extends $W/2$ both above and below the conductor's surface. Let S_1 be the top cap, S_2 the curved part above the conductor, S_3 the curved part inside the conductor, and S_4 the bottom cap. We know $\oint_S \vec{E}\cdot d\vec{S} = \oint_{S_1} \vec{E}\cdot d\vec{S} + \oint_{S_2} \vec{E}\cdot d\vec{S} + \oint_{S_3} \vec{E}\cdot d\vec{S} + \oint_{S_4} \vec{E}\cdot d\vec{S}$, but $\vec{E} = \vec{0}$ inside the conductor, so the last two integrals are zero. But $\oint_{S_2} \vec{E}\cdot d\vec{S} = 0$ as well: the unit normal to S_2 points away from



the cylinder's axis, but this axis is in the same direction as $\vec{E}$, so $\vec{E}\cdot d\vec{S} = 0$ on S_2.

This gives $\oint_S \vec{E}\cdot d\vec{S} = \oint_{S_1} \vec{E}\cdot d\vec{S}$. The normal to S_1 points



in the z-direction, so $\vec{E}\cdot d\vec{S} = E(z)|d\vec{S}|$ on S_1. ~~The~~ Since S_1 is at a fixed value of z, $E(z)$ is constant on S_1, and so

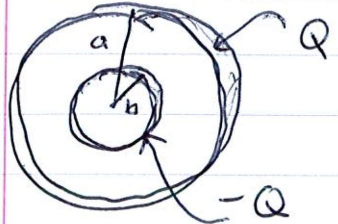
$$\oint_S \vec{E}\cdot d\vec{S} = \oint_{S_1} \vec{E}\cdot d\vec{S} = E(z)\oint_{S_1} |d\vec{S}| = E(z)A$$

since S_1 has area A. Therefore, we know $E(z)A = Q_S/\epsilon_0$. How much charge is contained in S? It intersects the conductor's surface on a circle of area A, so the total charge contained within is (surface density) $\times$ (area) $= \sigma A$. So, we find $E(z)A = \sigma A/\epsilon_0$, or $E(z) = \sigma/\epsilon_0$. But note the right-hand side is independent of z, so $\vec{E}$ is constant. Putting back in the direction thus gives the final answer:

$\boxed{\vec{E} = \frac{\sigma}{\epsilon_0}\hat{k}}$ above the conductor.

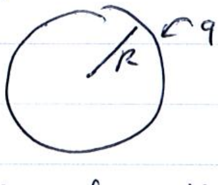
P.4

A capacitor is a device that can store electric charge, and it does so by constructing a configuration of conductors and then applying a voltage to it. We saw a couple of examples in lecture; now for another, a spherical capacitor.



So we have the configuration shown to the left. To find the capacitance, we need to calculate the difference in the potential between the two shells, i.e. $\Phi(a) - \Phi(b) = \Delta\Phi$.

But what's Φ? We know ~~this by~~ ~~the fact~~ that ~~the potential~~ ~~attached~~ to each shell, ~~but what's~~ this potential? Let's see: consider a single spherical shell of radius R and uniform surface charge q. If S is a spherical surface of radius r centred on the sphere, then



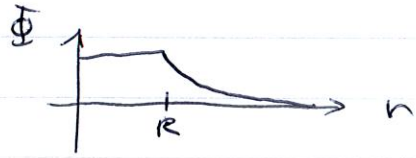
the same argument we've used several times before gives $\oint_S \vec{E}\cdot d\vec{S} = 4\pi r^2 E(r)$, since $\vec{E} = E(r)\hat{r}$ is the only possible form of the electric field. Thus, $E(r) = \frac{Q_S}{4\pi\epsilon_0 r^2}$. But if $r<R$, then $Q_S = 0$, since it's a hollow shell. Thus, $\vec{E} = \vec{0}$ inside a shell of uniformly-distributed charge! If $r>0$, then all the charge is inside S, so $E(r) = \frac{q}{4\pi\epsilon_0 r^2}$, i.e. $\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q\vec{r}}{r^3}$.

So here's a basic truth: inside a uniformly-charged spherical shell, the electric field is zero, and outside, it's the same as that due to a point charge.

~~But what's the potential?~~ We know that $\vec{E} = \frac{1}{4\pi\epsilon_0}\frac{q\vec{r}}{r^3}$ is obtained from a potential $\Phi = \frac{1}{4\pi\epsilon_0}\frac{q}{r}$, so that's the potential outside the shell. Inside, though, $\vec{E} = \vec{0} \Rightarrow \vec{\nabla}\Phi = \vec{0} \Rightarrow \Phi = \text{const}$.

So the potential Φ inside a spherical shell is a constant. If we want Φ to be continuous, this constant must be the value of $\frac{1}{4\pi\epsilon_0}\frac{q}{r}$ at $r=R$, so overall,

$$\Phi = \begin{cases} \frac{1}{4\pi\epsilon_0}\frac{q}{R} & r<R \\ \frac{1}{4\pi\epsilon_0}\frac{q}{r} & r>R \end{cases}$$



Now, back to our capacitor: the potential due to the inner shell is

$$\Phi_{\text{inner}} = \begin{cases} -\frac{1}{4\pi\epsilon_0}\frac{Q}{b} & r<b \\ -\frac{1}{4\pi\epsilon_0}\frac{Q}{r} & r>b \end{cases}$$

and for the outer shell, it's

$$\Phi_{\text{outer}} = \begin{cases} \frac{1}{4\pi\epsilon_0}\frac{Q}{a} & r<a \\ \frac{1}{4\pi\epsilon_0}\frac{Q}{r} & r>a \end{cases}.$$

To find $\Delta\Phi$, we need the total potential between the shells, i.e. in the region $b < r < a$. This is

$$\Phi = \Phi_{inner} + \Phi_{outer} = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{r}\right) \qquad b < r < a.$$

Thus, $\Phi(a) = 0$ and $\Phi(b) = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right)$. The potential difference is thus $\Delta\Phi = \Phi(a) - \Phi(b) = \frac{Q}{4\pi\epsilon_0}\left(\frac{1}{b} - \frac{1}{a}\right)$. The capacitance is $C = Q/\Delta\Phi$, or

$$C = \frac{Q}{\Delta\Phi} = \frac{Q}{\frac{Q}{4\pi\epsilon_0}\left(\frac{1}{b} - \frac{1}{a}\right)} = \frac{4\pi\epsilon_0}{\frac{a-b}{ab}} = \boxed{\frac{4\pi\epsilon_0 ab}{a-b}}.$$

This is an interesting formula, because it indicates that you can make C large not only by making a & b both large, but also by making the gap between the spheres very small. For example, if $a = 1.01$ cm and $b = 1$ cm (i.e. a gap of 0.1 mm), we get

$$C = \frac{4\pi(8.85\times10^{-12}\,C^2 \cdot m^{-2} \cdot N^{-1})(1.01\times10^{-2}\,m)(1\times10^{-2}\,m)}{(1\times10^{-4}\,m)} = 1.12\times10^{-10}\,F$$

which looks small but recall that a farad is a huge amount of capacitance. Spherical capacitors aren't common, but they do exist.